

- 1.0 COURSE NUMBER: IE 214
- 2.0 COURSE TITLE: Introductory Operations Research
- 3.0 COURSE DESCRIPTION: Operations research methodology. Basic linear programming, network analysis. Duality and sensitivity analysis.
- 4.0 PREREQUISITE:
- 5.0 CREDIT: 3.0 Units
- 6.0 COURSE OBJECTIVES:

At the end of the course, the student is expected to be able to accomplish the following:

- a) The student must be able to apply the OR methodology in problem solving;
- b) The student must be knowledgeable in the basic concepts, principles and techniques in linear programming (LP), sensitivity analysis and selected network models;
- c) The student must be able to identify opportunities for the application of linear programming for improved decision-making.
- d) The student must be familiar with the basic functions in modeling linear programming models in OR optimization software.

7.0 COURSE OUTLINE:

Week	Topic
1	Introduction to Operations Research, OR Methodology
2	Introduction to Linear Programming, LP Model Formulation
2	Graphical Solution
3	Basic Feasible Solution, Simplex Method
3	Big-M Method
4	Two-Phase Method, Special Cases of LPs
5	Problem Solving Exercises
6	Dual Problem, Primal-Dual Relationships, Complementary Slackness
6	Economic Interpretation of Duality
7	Dual Simplex Method
7	Simplex Tableau Expressed in terms of LP Parameters
8	Midterm Exam
9	Sensitivity Analysis: Changes in RHS, Objective Function and Technological Coefficients, Addition of New Activity, Addition of a New Constraint
10	Transportation Models, Starting Basic Feasible Solutions
10	Transportation Algorithm, Method of Multipliers
11	Assignment Model
11	Transshipment Problem
12	Problem Solving Exercises
13	Network Models, Terminologies, Minimal Spanning Tree Algorithm
13	Shortest Route Model, Shortest Route Algorithm,
14	Maximal Flow Models, Maximal Flow Algorithms, Enumeration of Cuts,
15	Problem Solving Exercises
16	Modeling LPs in a Software Environment and Demonstration of Software Functions

8.0 REFERENCES:

Taha, Hamdy, Operations Research, An Introduction, Pearson Education, 9th edition, 2011. ISBN-10: 0-13-139199-2.

Hillier, Frederick and Lieberman, Gerald, Introduction to Operations Research, McGraw-Hill, 9th edition, 2009, ISBN-10: 0077298349.

9.0 REQUIREMENTS

There are three requirements namely: Midterm Exam, Final Exam and Machine Problem (MP). The student's class standing will be calculated as follows:

Class Standing Calculation = $0.35 \times \text{Midterm Exam} + 0.40 \times \text{Final Exam} + 0.25 \times \text{MP}$

If a requirement is missed due to a valid concern and approved by the instructor, a make-up requirement will then be given at a later date, else it will be given a grade of zero. Request for the make-up requirement (with documentary evidence) must be submitted one week from the exam/requirement date at the latest.

Class Standing Grade Equivalent

Percent Range	Grade Equivalent
90 and above	1.00
86 to 90	1.25
82 to 86	1.50
78 to 82	1.75
74 to 78	2.00
71 to 74	2.25
68 to 71	2.50
64 to 68	2.75
55 to 64	3.00
Below 55	5.00

Lowell Lorenzo
email: Lowell.Lorenzo@up.edu.ph
Phone: 8981-8500 ext 3149/3128